
title: “Social Intelligence as a Field Phenomenon: Rapport, Empathy, and Collective Regulation Under the Universal Somatic Field” subtitle: “Fractal Programme — Social Scale (9–10)” author: “Alistair Johnson” orcid: “0009-0007-2194-0850” institute: “Independent Researcher, Zurich, Switzerland” date: “2026” lang: en-GB bibliography: ../../paper/bibliography.bib csl: ../../paper/apa-7th.csl abstract: | We apply the Universal Somatic Field (USF) framework to the social scale (scales 9–10, 10^1 – 10^3 m), deriving a formal mathematical model of social intelligence. The key identification: rapport between two people is Huygens frequency locking between their respective soma-fields, governed by the dyadic propagator $G_{AB}(\lambda)$ whose poles are the shared attractor modes accessible to both parties through coupling. Social intelligence quotient (SQ) is defined as the spectral gap of this dyadic propagator — the distance between the shared attractor and the next unstable mode. High SQ corresponds to a wide Arnold tongue: a large coupling bandwidth over which frequency locking is stable. Low SQ (as in social anxiety or autistic social processing differences) corresponds to a narrow Arnold tongue requiring precise frequency matching that cannot tolerate perturbation. The model makes three testable predictions: (1) interpersonal synchrony metrics correlate with dyadic propagator pole spacing; (2) therapeutic alliance quality predicts tunnelling amplitude across the social barrier; (3) group cohesion in organisations follows $O(N^2)$ scaling via the swarm propagator rather than $O(N \cdot K)$ message-passing. All structural claims are formally verified in the companion Lean 4 file DyadicField.lean.